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GENERAL MANAGER

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May 24, 2023

Michael Regan, Administrator
Environmental Protection Agency
1200 Pennsylvania Avenue, N.W.
Mail Code: 1309
Washington, DC 20004

RE: Per- and polyfluoroalkyl substances (PFAS): Perfluorooctanoic acid (PFOA) and
Perfluorooctanesulfonic acid (PFOS) National Primary Drinking Water Regulation Rulemaking,
Docket No. [EPA-HQ-OW-2022-0114](#)

Dear Administrator Regan,

I am the General Manager for Lakewood Water District (LWD) submitting comments on EPA's proposed PFAS drinking water regulations. LWD is the largest water District in Pierce County, Washington. Located just south of Tacoma, Washington, and adjacent to Joint Base Lewis McChord (JBLM). LWD serves more than 65,000 customers directly and over 60,000 more via wholesale water supply contracts with five neighboring purveyors.

LWD has been dealing with PFAS contamination of its aquifer system since 2016 when PFAS migrating from JBLM was first detected in the District's wells. PFAS has now been detected in 13 of the District's wells resulting in four of those wells being shut down due to exceedance of Washington State PFAS State Action Levels (SAL).

To address this widespread PFAS contamination, LWD has developed a multi-year \$28 million program to clean up and mitigate PFAS contamination at the 13 contaminated wellfields. So far, this program has cleaned up two wellfields (four wells) utilizing Granular Activated Carbon (GAC) systems at the cost of more than \$8.86 million. Supplementing this treatment approach, LWD is also pursuing drilling replacement wells into much deeper (hopefully) uncontaminated aquifers. These new wells are expected to offset supplies lost due to PFAS contamination.

With this context in mind, LWD's comments on EPA's proposed PFAS rules are framed from actual real-world experience with addressing PFAS contamination of drinking water. Our comments fall into six broad categories:

1. Costs are systematically underestimated.
2. Federal funding support is exaggerated.
3. Proposed trigger levels are inappropriate.
4. Proposed Hazard Index approach is flawed.
5. Penalties for monitoring are counterproductive.
6. Timeline to implement is not feasible.

Costs are Systematically Underestimated

The cost presented in the proposed rules systematically underestimates:

- ✓ Capital costs
- ✓ Operational costs, and
- ✓ Implementation costs

The following paragraphs provide examples and highlights regarding this underestimation and are not intended to be an exhaustive treatise. Rather, these examples highlight the serious shortcomings in the EPA's assumptions regarding the cost of implementation and compliance with the proposed rules.

Capital Costs

As described above, LWD has completed two GAC systems to remove PFAS from groundwater. LWD's first GAC system was at the Ponders well site, which has two wells. Design started in April 2018; the GAC system came online in January 2020. The all-in costs, including engineering, permitting, bidding, and administrative costs, were \$ 3.3 million for just over 2,000 gallons per minute (GPM), or \$1.65 million per 1,000 GPM.

LWD's second site, Scott's well site, has two wells producing about 3,000 GPM, with all the same components mentioned above. LWD started design in May 2020, completed construction, and brought it online in May 2022 for a cost of \$5.1 million for treating 3,000 GPM. Or \$1.70 million per 1,000 GPM.

The bottom line is that the capital costs that LWD has experienced are three times the cost of what EPA portrays in the proposed rulemaking.

Further, EPA's cost estimates for drilling new wells dramatically underestimate the costs. EPA estimates the cost of bringing a new well online to be less than \$1,000,000. As described previously, the LWD PFAS mitigation program also includes drilling new wells to replace contaminated wells. LWD's actual experience with the cost of drilling and bringing online one new well is closer to \$3.8 million. Keep in mind that the hydrogeology for LWD is more favorable for developing new wells. Nevertheless, LWD's actual experience with the cost of new wells is nearly four times the cost estimate presented by EPA in the proposed rules.

While these data points may be considered anecdotal, the magnitude of these examples of capital cost under-estimating is consistent with the broader findings of the American Water Works Association (AWWA) assessment of cost under-estimating in the proposed rule. The AWWA assessment also found that the typical PFAS treatment system is shown to cost 330% more than the estimated cost by the EPA's WBS Model.

Operational Costs

Similar to capital cost estimates, operational cost estimates consistently underestimate the operational cost impact that water purveyors will face. LWD's annual Operation and Maintenance (O&M) cost for staff time, site visits, and sampling costs is \$60,000 for two GAC treatment sites. Under the current Washington State SAL level of 10 parts per trillion (ppt) for PFOA and 15 ppt for PFOS, the estimated run time for replacing media is \$400,000 per site every five years. With the proposed EPA MCL of 4 ppt for PFOS and PFOA, LWD would have to replace the media every 2.5 years, which would double those costs.

Implementation Costs

We were struck by the dramatic underestimation of the level of effort needed to stand up the systems and organizations needed to implement these proposed PFAS regulations. **Table 36** is a standout example of this underestimation.

Table 36 suggests that a Primacy Agency (such as we have here in Washington State) could “...read and understand the rule, as well as adopt regulatory requirements” (emphasis added) with 416 hours of staff time. This estimate is so low that it must be a clerical error.

In order to adopt regulatory requirements, most states (like Washington state) have specific and very detailed administrative procedures that must be adhered to in order to adopt water quality regulations. The required administrative procedures at the state level are not unlike the procedures that the EPA is following for the development of its PFAS rules. Rulemaking at the state level takes months and requires massive amounts of specialized staff time, many times the 416 hours shown in **Table 36**.

Table 36 also suggests that a Primary Agency could “...provide initial training and technical assistance to systems” with 2080 hours of staff time. 2080 hours is one Full Time Equivalent employee. Washington State (which is not unique in this regard) has over 2,400 water systems that will be covered by this proposed rule. 2080 hours means less than 1 hour of “training and technical assistance” would be provided per system. This is totally unrealistic.

These are but two examples of gross underestimation of implementation costs. EPA must revisit these cost estimates to update them to more realistic numbers.

Funding Support Exaggerated

The proposed regulations highlight the \$11.7 billion of Drinking Water SRF funding contained in the Bipartisan Infrastructure Law (BIL). \$4 billion of this appropriation is set aside for Emerging Contaminants like PFAS. While \$4 billion is certainly a lot of money, the implication that this will largely offset the cost of compliance is misleading.

Table 66 shows that the annual cost of compliance with these proposed rules ranges from \$705 million (on the optimistic side) to over \$1.3 billion (on the more realistic side). At these rates, the BIL support will last at most about five years and, more likely, about three years.

The description of federal dollars fails to mention that this support could often be in the form of loans and is, therefore, not really funding but rather a financing option.

Further, whether federal dollars come in the form of a grant, forgivable loan, or loan, all federal money comes with multiple regulatory strings attached that increase costs (e.g., Buy America Build America requirements, American Iron and Steel requirements, recycled material requirements, disadvantaged business requirements, and more).

In addition to the imposition of additional costs, securing federal dollars takes an exceptionally long time and almost always lengthens the duration of a project, which also increases costs and diminishes the value of federal support.

Trigger Level Inappropriate

The proposed regulation clearly documents that the minimum Practical Quantitative Limit (PQL) for PFOA and PFOS is 4 ppt. Contrary to this fact, EPA proposes using a value of one-third of the PQL to trigger monitoring requirements for water systems. It is inappropriate and unreasonable to base regulatory decisions on data that, by definition, is unreliable.

Further, using two different regulatory levels will create massive risk communication challenges for water purveyors like LWD. No values below PQL should be used for regulatory purposes.

Hazard Index Approach is Flawed

The proposed Hazard Index (HI) approach is fundamentally flawed. Under the proposed HI approach, a water system could document that all four constituents of the Hazard Index are below the Health Based Water Concentrations (HBWCs) and still be characterized as hazardous. In fact, EPA shared this very scenario during several webinars with Example 2 of the HI calculations.

In EPA's Example 2, all four constituents are below the HBWC, yet the result was a HI above 1. This does not make sense. The HI concept should be significantly revised or discarded.

Monitoring Penalties Counterproductive

As proposed, the regulations would seem to penalize water purveyors like LWD that have proactively and aggressively addressed PFAS contamination of their systems. LWD has been monitoring its wells for PFAS and reporting those results for many years. Under EPA's proposed rules, LWD now runs the risk of being classified (once EPA's MCLs are adopted) in violation of PFAS MCLs even though the regulatory framework in our state (Washington) will not have had time to be stood up.

EPA, instead, should postpone enforcement of PFAS MCLs until after state compliance systems and programs have been formally established.

Timeline to Implement

The three-year timeline proposed to complete 12 months of sampling and then to put any mitigation steps into place is not feasible. With recent experience, an aggressive design schedule for a treatment facility is 12 months.

In addition, many regulatory agencies will require pilot testing for treatment systems, which will further add to the time required to implement system improvements. With many jurisdictions and engineering firms being understaffed, the permitting and design schedule can extend significantly. A typical bid advertisement and award timeframe can easily be two months. It is not unreasonable to expect an 18-month timeframe before construction is ready to begin. This is before any treatment equipment is ordered as well. Currently, we see about a 6–12-month lead time for pumps and upwards of 12 months lead time for pressure filter vessels used in our GAC system. In addition, it would be safe to assume other utilities across the United States would also select GAC systems which will undoubtedly increase lead times.

We have seen similarly long lead times for GAC media for new installations. Even with 12 months of sampling data, the design, bidding, equipment procurement, and construction may take up to three years under typical practices. This is even before the anticipated significant increase in demand for engineering, equipment, and construction created by water systems throughout the country, requiring goods from a

limited pool of qualified engineers, manufacturers, and construction contractors. A timeline of at least five years would still be an aggressive schedule but at least somewhat more realistic.

Funding timelines further exacerbate the ability to meet the proposed timeline for the rule. As an example, LWD was fortunate to receive direct appropriations to help offset costs for two new wells through Senator Murray's office in May 2021 and still has not been able to receive a contract from the EPA program charged with administering the funding. With the federal funds described in the rule, the timeline for systems fortunate to receive funds is further extended past the proposed rule's effective date.

Sincerely,

A handwritten signature in black ink, reading "Randall M. Black". The signature is written in a cursive, flowing style.

Randall M. Black
General Manager